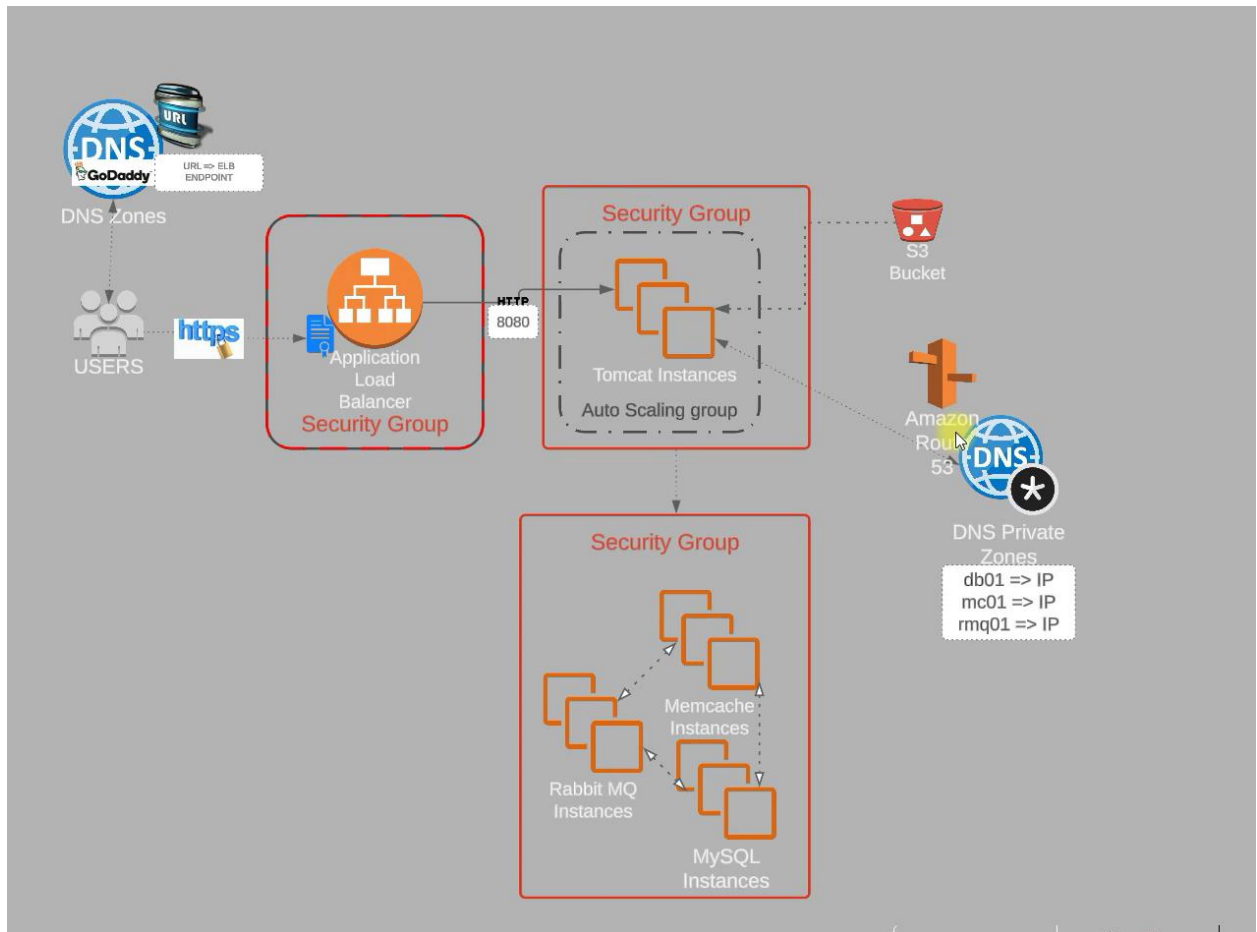


LIFT AND SHIFT APPLICATION WORKLOAD

APP ARCHITECTURE:



STEP 1: Create key pairs

Key pairs (1/1) [Info](#)

[Actions](#) [Create key pair](#)

☒	Name	Type	Created	Fingerprint	ID
☒	vprofile-prod-key	rsa	2025/01/09 00:04 GMT-5	ef:37:48:a4:07:99:33:1e:06:65:c3:88:83:79:b8:7f:ca:...	key-0e73292af94daac3a

STEP 2: Create security group

sg-0e58ebb7c5218eff2 - vprofile-backend-sg

Actions

Details

Security group name
vprofile-backend-sg

Security group ID
sg-0e58ebb7c5218eff2

Description
SG for mysql, memecache & rabbitmq allowed from tomcat app server

VPC ID
vpc-0c4b2764c716015a4

Owner
011528261245

Inbound rules count
6 Permission entries

Outbound rules count
1 Permission entry

Inbound rules

Outbound rules

Sharing - new

VPC associations - new

Tags

Inbound rules (6)



Manage tags

Edit inbound rules

Search

< 1 > ⚙

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
☐	-	sgr-0160c74aae7dbeac9	IPv4	All ICMP - IPv4	ICMP	All	0.0.0.0/0	-
☐	-	sgr-0bf11c4c39d00cf4a	IPv4	SSH	TCP	22	142.188.196.35/32	-
☐	-	sgr-0a40aa0d8826b6098	-	Custom TCP	TCP	11211	sg-01f31e8ac2c3f1b8b...	-
☐	-	sgr-068c734ad40c533b9	-	All traffic	All	All	sg-0e58ebb7c5218eff2...	-
☐	-	sgr-02fafe8feb3550514	-	MYSQL/Aurora	TCP	3306	sg-01f31e8ac2c3f1b8b...	-
☐	-	sgr-0bb4cf09b101a5ea8	-	Custom TCP	TCP	5672	sg-01f31e8ac2c3f1b8b...	-

sg-01f31e8ac2c3f1b8b - vprofile-app-sg

Actions

Details

Security group name
vprofile-app-sg

Security group ID
sg-01f31e8ac2c3f1b8b

Description
Security Group for tomcat app server

VPC ID
vpc-0c4b2764c716015a4

Owner
011528261245

Inbound rules count
3 Permission entries

Outbound rules count
1 Permission entry

Inbound rules

Outbound rules

Sharing - new

VPC associations - new

Tags

Inbound rules (3)



Manage tags

Edit inbound rules

Search

< 1 > ⚙

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
☐	-	sgr-0e22ac0b90e5384bc	IPv4	Custom TCP	TCP	8080	142.188.196.35/32	-
☐	-	sgr-09b181e842fec0adf	IPv4	SSH	TCP	22	142.188.196.35/32	-
☐	-	sgr-04c25bc30e9a01a0a	-	Custom TCP	TCP	8080	sg-0b55dec92df156b484...	Allows traffic from

sg-0b55dec92df156b484 - vprofile-ELB-SG

Actions

Details

Security group name
vprofile-ELB-SG

Security group ID
sg-0b55dec92df156b484

Description
Security Group for the vprofile load balancer

VPC ID
vpc-0c4b2764c716015a4

Owner
011528261245

Inbound rules count
4 Permission entries

Outbound rules count
1 Permission entry

Inbound rules

Outbound rules

Sharing - new

VPC associations - new

Tags

Inbound rules (4)



Manage tags

Edit inbound rules

Search

< 1 > ⚙

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
☐	-	sgr-038eb67debebf3986	IPv4	HTTPS	TCP	443	0.0.0.0/0	-
☐	-	sgr-0beab2389f1a59fa4	IPv6	HTTPS	TCP	443	::/0	-
☐	-	sgr-0153d903c268399ab	IPv6	HTTP	TCP	80	::/0	-
☐	-	sgr-013cf2242dd2beb37	IPv4	HTTP	TCP	80	0.0.0.0/0	-

STEP 3: Launch Instances with user data (BASHSCRIPTS)

Instances (5) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
☐	vprofile-rmq01	i-03bda3f24bfe033cb	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	ec2-44-204-35-58.com...	44.204.35.58	-
☐	vprofile-app01	i-063676801c9e0b405	Terminated	t2.micro	-	View alarms +	us-east-1b	-	-	-
☐	vprofile-db01	i-026a7d025050a64f0	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1b	ec2-3-80-119-237.com...	3.80.119.237	-
☐	vprofile-mc01	i-02be1ef88b9df94ff	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1b	ec2-3-95-187-204.com...	3.95.187.204	-
☐	vprofile-app	i-0a96875bbe930fed	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1f	ec2-3-237-7-196.comp...	3.237.7.196	-

STEP 4: Update IP to NAME mapping in route 53

Hosted zones (1)

Automatic mode is the current search behavior optimized for best filter results. To change modes go to settings.

Filter records by property or value

< 1 >

Hosted zone name	Type	Created by	Record count	Description
vprofile.in	Private	Route 53	6	-

Private vprofile.in Info

Delete zone

Test record

Configure query logging

Hosted zone details

Edit hosted zone

Records (6)

Hosted zone tags (0)

Records (6) Info

Automatic mode is the current search behavior optimized for best filter results. To change modes go to settings.

Filter records by property or value

Type

Routing p...

Alias

< 1 >

☐	Record name	Type	Routin...	Differ...	Alias	Value/Route traffic to	TTL (s...
☐	vprofile.in	NS	Simple	-	No	ns-1536.awsdns-00.co.uk. ns-0.awsdns-00.com. ns-1024.awsdns-00.org. ns-512.awsdns-00.net.	172800
☐	vprofile.in	SOA	Simple	-	No	ns-1536.awsdns-00.co.uk. a...	900
☐	app01.vprofile.in	A	Simple	-	No	172.31.17.198	300
☐	db01.vprofile.in	A	Simple	-	No	172.31.20.85	300
☐	mc01.vprofile.in	A	Simple	-	No	172.31.18.98	300
☐	rmq01.vprofile.in	A	Simple	-	No	172.31.89.255	300

STEP 5: Build application from source code

STEP 6: Upload to s3 bucket

```
ajusi@Arjun-PC MINGW64 /e/hkhcoder/vprofile-project (awsliftandshift)
$ aws configure
AWS Access Key ID [None]: AKIAQFLZDJZ64KF2FD7I
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]: json
ajusi@Arjun-PC MINGW64 /e/hkhcoder/vprofile-project (awsliftandshift)
$ ls target/
classes/          maven-archiver/  test-classes/
generated-sources/ maven-status/    vprofile-v2/
generated-test-sources/ surefire-reports/ vprofile-v2.war

ajusi@Arjun-PC MINGW64 /e/hkhcoder/vprofile-project (awsliftandshift)
$ aws s3 cp target/vprofile-v2.war s3://vprofile-las-artifacts3
upload: target\vprofile-v2.war to s3://vprofile-las-artifacts3\vprofile-v2.war

ajusi@Arjun-PC MINGW64 /e/hkhcoder/vprofile-project (awsliftandshift)
$
```

S3 bucket

vprofile-las-artifacts3 [Info](#)

[Objects](#) | [Properties](#) | [Permissions](#) | [Metrics](#) | [Management](#) | [Access Points](#)

Objects (1) [Info](#)

[Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

< 1 > [Settings](#)

☐	Name	Type	Last modified	Size	Storage class
☐	vprofile-v2.war	war	January 9, 2025, 02:00:31 (UTC-05:00)	79.4 MB	Standard

STEP 7: Download artifact to tomcat EC2 instance

STEP 8: Setup ELB WITH http and https if domain name is available

OTHER AWS CONFIGURATION

LAUNCH TEMPLATE

Launch Templates (1) [Info](#)

Q Search

< 1 >

☐	Launch Template ID	Launch Template Name	Default Version	Latest Version	Create Time	Created By	Managed	Operator
☐	lt-09b338577e7fed62c	vprofile-las-app-lt	1	1	2025-01-09T07:25:09.000Z	arn:aws:iam::011528261245:root	false	-

AMI

Amazon Machine Images (AMIs) (1) [Info](#)

Owned by me

Q Find AMI by attribute or tag

Recycle Bin

EC2 Image Builder

Actions

Launch instance from AMI

< 1 >

☐	Name	AMI name	AMI ID	Source	Owner	Visibility	Status	Creation date
☐	vprofile-las-image-ami		ami-09bbf5ac356d04ca9	011528261245/vprofile-las-image-ami	011528261245	Private	Available	2025/01/09 02:22 GMT-5

Select an AMI

LOAD BALANCERS

Load balancers (1)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

< 1 >

☐	Name	DNS name	State	VPC ID	Availability Zones	Type	Date created
☐	vprofile-las-elb	vprofile-las-elb-214105285...	Active	vpc-0c4b2764c716015a4	5 Availability Zones	application	January 9, 2025, 02:16 (UTC-05:00)

0 load balancers selected

Select a load balancer above.

Listeners and rules (1) [Info](#)

Manage rules

Manage listener

Add listener

A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action and any additional rules.

Filter listeners

< 1 >

☐	Protocol:Port	Default action	Rules	ARN	Security policy	Default SSL/TLS certificate
☐	HTTP:80	Forward to target group - vprofile-las-tg 1 (100%) - Target group stickiness: Off	1 rule	ARN	Not applicable	Not applicable

vprofile-las-tg
Actions ▾

Details

arn:aws:elasticloadbalancing:us-east-1:011528261245:targetgroup/vprofile-las-tg/aec123abbf9d6ba8

Target type Instance	Protocol : Port HTTP: 8080	Protocol version HTTP1	VPC vpc-0c4b2764c716015a4
IP address type IPv4	Load balancer vprofile-las-elb		

1 Total targets	1 Healthy 0 Anomalous	0 Unhealthy	0 Unused	0 Initial	0 Draining
---------------------------	--	-----------------------	--------------------	---------------------	----------------------

► **Distribution of targets by Availability Zone (AZ)**

Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets |
 Monitoring |
 Health checks |
 Attributable |
 Tags

Registered targets (1) Info

Anomaly mitigation: Not applicable

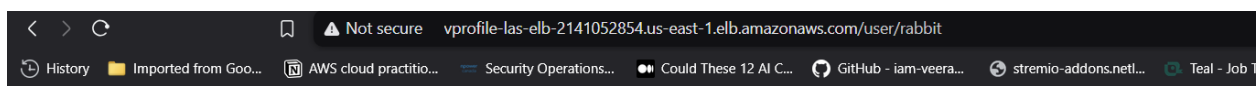
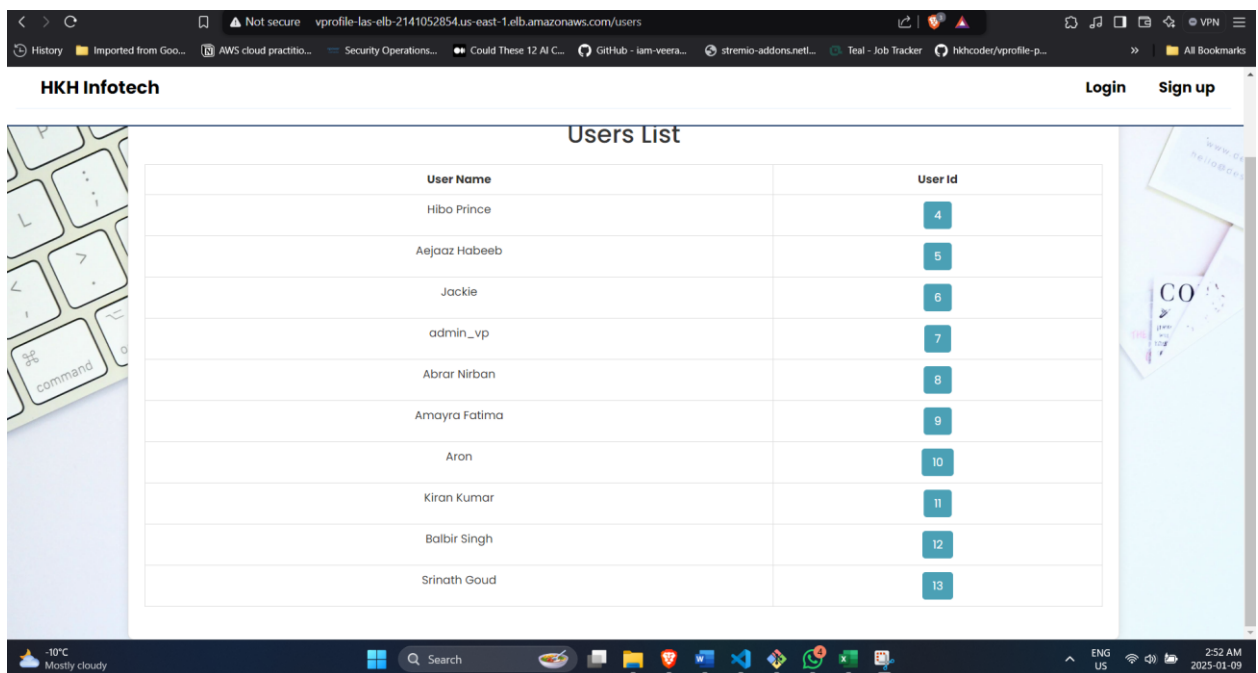
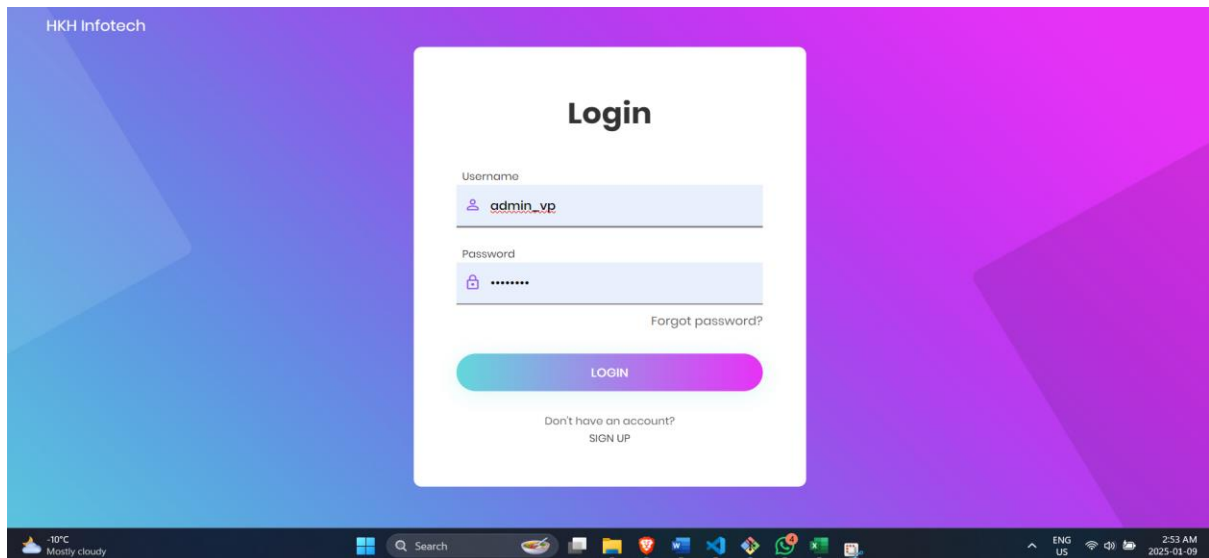
[Deregister] [Register targets]

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

< 1 >

☐	Instance ID	Name	Port	Zone	Health status	Health status details	Admini...	Overri...	Launch... ▲	An
☐	i-0a956875bbce930fed	vprofile-app	8080	us-east-1f(usc...)	Healthy	-	No override.	No overri...	January 9,...	

The screenshot shows a web browser window displaying a LinkedIn profile for HKH Infotech. The browser's address bar shows the URL "vprofile-las-elb-2141052854.us-east-1.elb.amazonaws.com/welcome". The LinkedIn header includes navigation links like "Home", "My Activity", and "Search". The profile section features a profile picture of a baby wearing glasses, the company name "HKH Infotech", and a bio describing their focus on DevOps for Product Management and Strategy of Application Delivery. Below the bio, there are statistics for posts, photos, and contacts. A post titled "The Key to DevOps Success" is visible, discussing collaboration in DevOps.



RabbitMQ Initiated

Generated 7 Connections

4 Channels, 7 Exchange, and 3 Queues